Lines and Planes in Three Dimensions

ClipSAT Printable Practice Suite (clipsat.org) • Topic 3.02

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: The vector equation of a line**

Write the vector equation of the line through the point $(2, -1, 3)$ with direction vector $(1, 2, -1)$.

2 **[Foundational] • Target: The vector equation of a line**

Find the vector equation of the line through the two points $A(1, 0, 2)$ and $B(4, 3, 1)$.



3 **[Foundational] • Target: Determining if two lines intersect**

Determine whether the lines $\mathbf{r}_1 = (1, 2, 0) + s(1, -1, 2)$ and $\mathbf{r}_2 = (0, 3, 1) + t(2, 1, -1)$ intersect, and if so, find the point of intersection.

4 **[Foundational] • Target: The equation of a plane**

Find the Cartesian equation of the plane with normal vector $(2, -1, 3)$ passing through the point $(1, 2, -1)$.

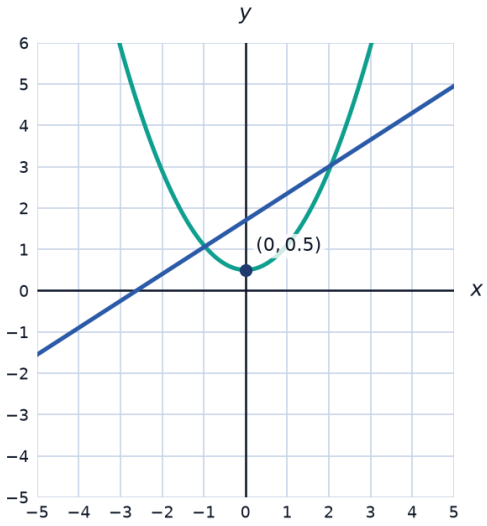
5 **[Foundational] • Target: The equation of a plane**

Find the Cartesian equation of the plane containing the points $A(1, 0, 0)$, $B(0, 1, 0)$, and $C(0, 0, 1)$.

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6 [Foundational] • Target: Angle between a line and a plane

Find the angle between the line with direction $(1, 2, 2)$ and the plane $x + y + z = 5$, to the nearest degree.



7 [Foundational] • Target: Distance from a point to a plane

Find the shortest distance from the point $(3, 1, -2)$ to the plane $2x - y + 2z = 6$, using $d = \frac{|ax_0 + by_0 + cz_0 - k|}{\sqrt{a^2 + b^2 + c^2}}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Finding the angle between two planes

Find the angle between the planes $x + 2y - z = 3$ and $2x - y + z = 5$, to the nearest degree.

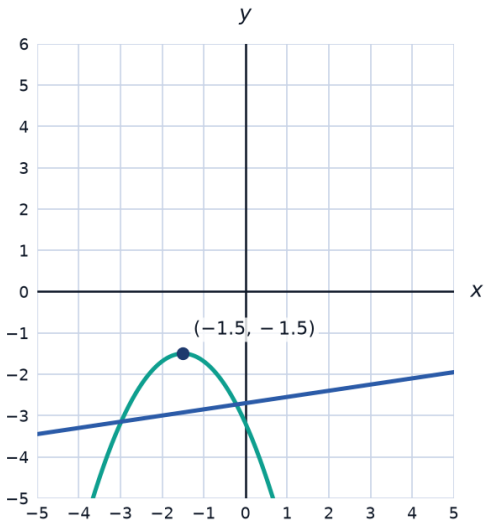
9 [Intermediate] • Target: Finding the intersection line of two planes

Find a vector equation for the line of intersection of the planes $x + y + z = 6$ and $x - y + 2z = 5$, using the cross product of the two normal vectors as the direction, and finding one point on both planes by inspection.

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10 [Intermediate] • Target: Testing whether a line lies within a plane

Determine whether the line $\mathbf{r} = (1, 2, 3) + t(2, -1, 1)$ lies entirely within the plane $x + 3y + z = 10$.



11 [Intermediate] • Target: Finding the point where a line meets a plane

Find the point where the line $\mathbf{r} = (1, 0, 2) + t(1, 2, -1)$ intersects the plane $2x + y - z = 3$.

12 [Intermediate] • Target: Shortest distance between two skew lines

Two skew lines have direction vectors $\mathbf{d}_1 = (1, 0, 1)$ and $\mathbf{d}_2 = (0, 1, 1)$, and pass through the points $A(0, 0, 0)$ and $B(1, 1, 0)$ respectively. Find the shortest distance between the two lines, using $d = |\vec{AB} \cdot (\mathbf{d}_1 \times \mathbf{d}_2)| / |\mathbf{d}_1 \times \mathbf{d}_2|$.

13 [Intermediate] • Target: Foot of the perpendicular from a point to a line

Find the foot of the perpendicular from the point $P(4, 0, 1)$ to the line $\mathbf{r} = (1, 0, 0) + t(1, 1, 1)$.

14 [Intermediate] • Target: Foot of the perpendicular from a point to a line

Determine whether the planes $2x - y + 3z = 4$ and $-4x + 2y - 6z = -8$ represent the same plane, a pair of parallel (distinct) planes, or intersecting planes.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

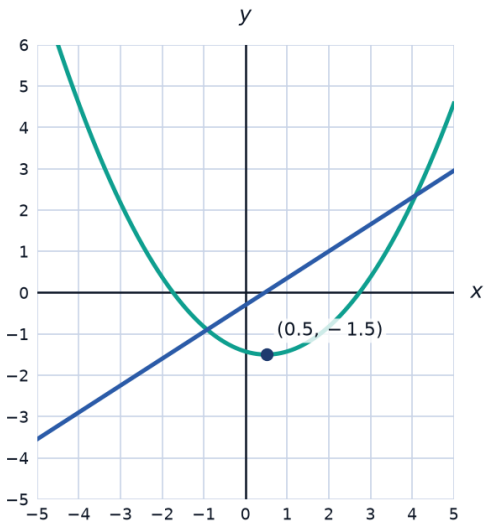
15 [Challenging] • Target: Checking whether a point lies on a plane • Cross-Lesson: 1.05 Systems of Linear Equations (AA HL)

Determine whether the point $P(2, -1, 4)$ lies on the plane $3x + 2y - z = 1$, and if not, state whether it lies on the same side as the origin.

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16 [Challenging] • Target: Checking whether a point lies on a plane • Cross-Lesson: 2.01 Modulus Functions & the Factor/Remainder Theorem

Find the value of k such that the line $\mathbf{r} = (1, 2, 0) + t(2, -1, k)$ is parallel to the plane $x + 2y - z = 5$.

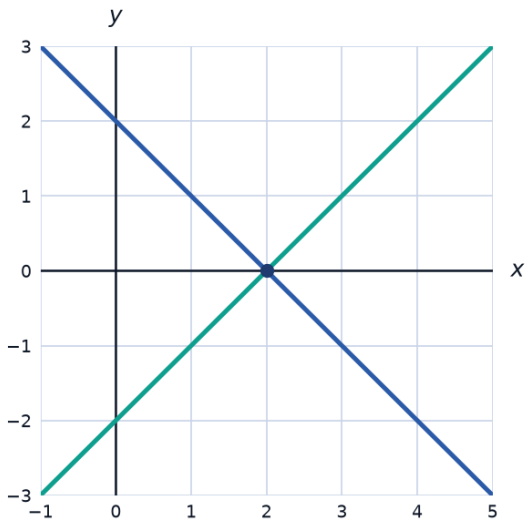


17 [Challenging] • Target: Checking whether a point lies on a plane • Cross-Lesson: 3.01 3D Vectors & the Cross Product

A surveyor models a hillside as the plane $2x + 3y + 6z = 42$, where x and y are horizontal coordinates (in metres) and z is height (in metres). Find the height z of the hillside at the point where $x = 6$ and $y = 4$, and find the plane's intercept on the z-axis (i.e. the height at $x = 0, y = 0$).

18 [Challenging] • Target: Checking whether a point lies on a plane • Cross-Lesson: 3.03 Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)

This question has three parts. Two lines are given by $\mathbf{r}_1 = (2, 0, 1) + s(1, 1, 0)$ and $\mathbf{r}_2 = (0, 2, 1) + t(1, -1, 0)$. (a) Show that both lines lie in the plane $z = 1$. (b) Find the point of intersection of the two lines. (c) Find the acute angle between the two lines, to the nearest degree.



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19 [Challenging] • Target: Converting between vector and Cartesian forms of a line • Cross-Lesson: 4.01 Unbiased Estimators & Confidence Intervals

Write the parametric equations of the line $\mathbf{r} = (2, 1, -3) + t(1, -2, 4)$, and hence give its Cartesian (symmetric) form.

20 [Challenging] • Target: Determining if two planes are parallel • Cross-Lesson: 4.02 The Poisson Distribution

Determine whether the planes $x - 2y + 3z = 5$ and $2x - 4y + 6z = 1$ are parallel, and if so, whether they represent the same plane.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: The vector equation of a line

Relative to a fixed origin O, the lines L_1 and L_2 have vector equations $L_1 : \mathbf{r} = (1; 0; -1) + \lambda(1; 2; 2)$, $L_2 : \mathbf{r} = (2; -4; 7) + \mu(2; 1; -2)$. The unique points P on L_1 and Q on L_2 are such that the line segment PQ is perpendicular to both L_1 and L_2. A point R lies on L_1 such that the area of triangle PQR is 18. Given that the z-coordinate of R is positive, what is the exact distance from the origin O to the midpoint of the line segment QR?

a. $3\sqrt{5}$

b. $\sqrt{13}$

c. $6\sqrt{2}$

d. $\sqrt{77}$

22 [Foundational] • Target: Determining if two lines intersect

The line l_1 has vector equation $\mathbf{r} = (1; 5; 0) + \lambda(1; -2; 2)$ and the line l_2 has vector equation $\mathbf{r} = (1; p; 2) + \mu(2; 3; q)$ where p and q are constants, and $\lambda, \mu \in \mathbb{R}$. Given that the lines l_1 and l_2 are perpendicular and intersect at a common point, find the value of p.

a. -2

b. 2

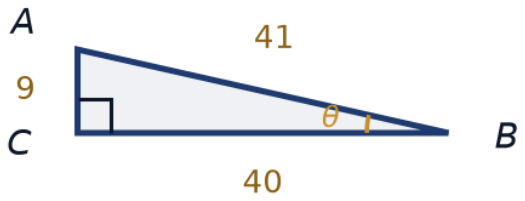
c. -4

d. 4

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23 [Foundational] • Target: The equation of a plane

A straight line passing through the origin is tangent to both the curve C_1 with equation $y = e^{kx}$, where k is a positive constant, and the curve C_2 with equation $y = \ln x$. Determine the exact distance between the two points of contact.



a. $(e-1)\sqrt{e^2+1}$

b. $e\sqrt{e^2+1}$

c. $(e-1)\sqrt{e^2-1}$

d. $\sqrt{e^4-e^2+1}$

24 [Foundational] • Target: Angle between a line and a plane

Consider a point $P(\cos\theta, \sin\theta)$ on the unit circle in the Cartesian plane. If the distance from the origin to the point $Q(x, y)$ is 5, and the angle between the positive x-axis and the line segment OQ is α, such that $\cos\theta = \frac{3}{5}$ and $\sin\theta = \frac{4}{5}$. If $x = \cos\alpha$ and $y = \sin\alpha$, find the maximum possible value of the product of the x-coordinates of P and Q, i.e., $\cos\theta \cdot \cos\alpha$.

a. 4/5

b. 3/5

c. 1

d. 5/3

25 [Foundational] • Target: Distance from a point to a plane

Two particles, A and B, move in the xy-plane. Their position vectors at time t seconds, for $t \geq 0$, are given by $\mathbf{r}_A(t) = (2t; 4t-1)$ and $\mathbf{r}_B(t) = (3+5t; 1-2t)$ respectively. The particles start their motion at $t = 0$. Find the minimum distance between particle A and particle B.

a. $\sqrt{26}$

b. $2\sqrt{13}$

c. $3\sqrt{10}$

d. $4\sqrt{5}$

26 [Foundational] • Target: Finding the angle between two planes

Consider two points in the Cartesian plane, $P = (2, 3)$ and $Q = (5, 7)$. What is the component form of the vector $\vec{PQ}$?

a. $\langle 3, 4\rangle$

b. $\langle -3, -4\rangle$

c. $\langle 7, 10\rangle$

d. $\langle 1, 4\rangle$

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27 [Foundational] • Target: Finding the intersection line of two planes

Line of intersection of two non-parallel planes: direction vector =

a. sum of normals

b. cross product of normals

c. difference of normals

d. dot product of normals

28 [Foundational] • Target: Testing whether a line lies within a plane

Two vectors are anti-parallel if angle between them is:

a. 0

b. $\pi/4$

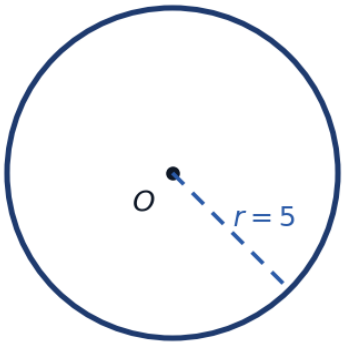
c. π

d. $\pi/2$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Finding the point where a line meets a plane

Consider the unit circle shown, where a point $P(x, y)$ lies on the circle in the first quadrant. A ray from the origin through P forms an angle θ with the positive x-axis. If the y-coordinate of P is $\frac{\sqrt{3}}{2}$ and $\cos\theta = \frac{1}{2}$, find the value of x.



a. $\frac{\sqrt{3}}{2}$

b. $\frac{1}{2}$

c. 1

d. $\frac{\sqrt{2}}{2}$

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30 **[Intermediate]** • **Target: Shortest distance between two skew lines**

The locus $|z-(1)| = |z-(4+3i)|$ is the perpendicular bisector of the segment joining which two points?

a. $(4, 3)$ and $(1, 0)$

b. $(1, 0)$ and $(4, 3)$

c. $(-1, 0)$ and $(-4, -3)$

d. $(1, 3)$ and $(4, 0)$

31 **[Intermediate]** • **Target: Foot of the perpendicular from a point to a line**

Consider a circle with the equation $x^2 + y^2 = 25$. A point $P(x_0, y_0)$ lies on this circle. A tangent line to the circle at point P intersects the x-axis at point A and the y-axis at point B. If the area of the triangle OAB, where O is the origin $(0, 0)$, is minimized, find the value of $x_0^2 + y_0^2$.

a. 25

b. 50

c. 100

d. 125

32 **[Intermediate]** • **Target: Checking whether a point lies on a plane**

A rectangular garden is to be enclosed by a fence. One side of the garden is adjacent to a long straight river, and requires no fencing. The other three sides require fencing, and the total length of the fence available is 120 meters. Let the side of the garden parallel to the river have length x meters, and let the two sides perpendicular to the river each have length y meters. The area of the garden is given by $A(x, y) = xy$. \\ Find the dimensions x and y that maximize the area of the garden.

a. x = 60 m, y = 30 m

b. x = 30 m, y = 60 m

c. x = 40 m, y = 40 m

d. x = 80 m, y = 20 m

33 **[Intermediate]** • **Target: Converting between vector and Cartesian forms of a line**

The diagram shows the graphs of the linear equation $y = 2x + c$ and the quadratic equation $y = -x^2 + 4x + 5$. The two graphs intersect at the points where $x = -1$ and $x = k$. Find the value of c and the value of k.

a. c = 3, k = 4

b. c = 5, k = 3

c. c = 3, k = 3

d. c = 5, k = 4

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34 [Intermediate] • Target: Determining if two planes are parallel

A particle of mass m kg is placed on a rough inclined plane. The plane is inclined at an angle α to the horizontal, where $\tan\alpha = \frac{3}{4}$. The particle is on the point of sliding down the plane. The coefficient of static friction between the particle and the plane is μ_s. Given that the acceleration due to gravity is $g = 9.8\ \text{m/s}^2$, find the value of μ_s.

a. $\frac{1}{2}$

b. $\frac{3}{4}$

c. $\frac{3}{5}$

d. $\frac{4}{5}$

35 [Intermediate] • Target: Distance between two parallel planes

Distance between skew lines $L_1 : \mathbf{r} = \mathbf{a} + s\mathbf{d}_1$ and $L_2 : \mathbf{r} = \mathbf{b} + t\mathbf{d}_2$:

a. $|\mathbf{b} - \mathbf{a}|$

b. $|(\mathbf{b} - \mathbf{a}) \cdot (\mathbf{d}_1 \times \mathbf{d}_2)|/|\mathbf{d}_1 \times \mathbf{d}_2|$

c. $|\mathbf{d}_1 - \mathbf{d}_2|$

d. 0

36 [Intermediate] • Target: Checking whether three points are collinear

A particle in equilibrium is acted on by three concurrent forces. Two of the forces have equal magnitude 20 N and act symmetrically either side of the third force's line of action. If the angle between each of the two equal forces and the third force is 120°, find the magnitude of the third force.

a. 18

b. 20

c. 25

d. 22

37 [Intermediate] • Target: Reflecting a point in a plane

Two planes are parallel if their normal vectors are:

a. parallel (scalar multiples)

b. perpendicular

c. equal in magnitude

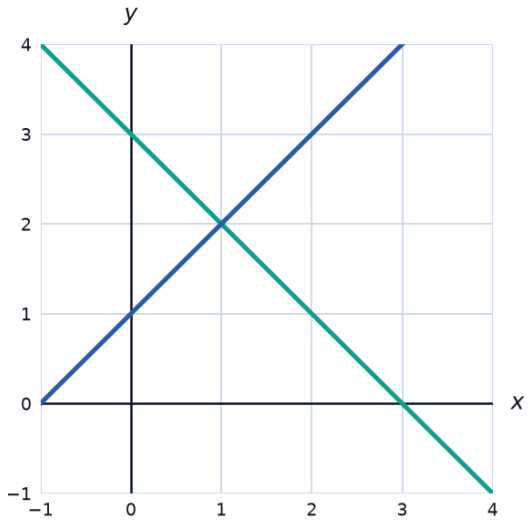
d. zero

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 **[Challenging]** • **Target: The vector equation of a line** • **Cross-Lesson: 3.03 Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)**

The diagram shows the graphs of two linear equations. The point of intersection of the two lines represents the solution to the system of simultaneous equations. Find the coordinates of the point of intersection.



a. (1, 2)

b. (2, 1)

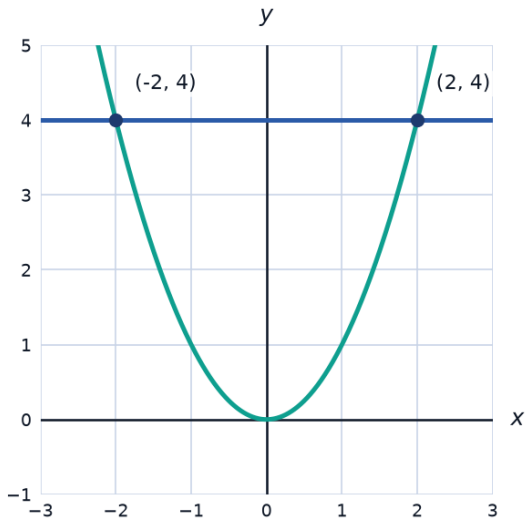
c. (3, 0)

d. (0, 3)

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39 **[Challenging]** • **Target: Determining if two lines intersect** • **Cross-Lesson: 4.01 Unbiased Estimators & Confidence Intervals**

Consider the region bounded by the curves $y = x^2$ and $y = 4$. This region is revolved around the line $y = 5$ to form a solid. A cylindrical hole is drilled through the center of this solid, perpendicular to the axis of revolution. The radius of the cylindrical hole is r. The volume of the material removed by the cylindrical hole is given by $V_{\text{hole}} = \pi\int_a^b (R(x)^2 - r^2)dx$, where $R(x)$ is the outer radius of the solid at a given x and a, b are the limits of integration. Find the value of a and b, and the expression for V_{hole} in terms of r. The question asks for the setup of the integral representing the volume of the material removed.



a. Limits of integration are -2 and 2; $V_{\text{hole}} = \pi\int_{-2}^{2}((5 - x^2)^2 - r^2)dx$

b. Limits of integration are 0 and 4; $V_{\text{hole}} = \pi\int_{0}^{4}((4 - x^2)^2 - r^2)dx$

c. Limits of integration are 1 and 4; $V_{\text{hole}} = \pi\int_{1}^{4}((5 - x^2)^2 - r^2)dx$

d. Limits of integration are -2 and 2; $V_{\text{hole}} = \pi\int_{-2}^{2}((4 - x^2)^2 - r^2)dx$

40 **[Challenging]** • **Target: The equation of a plane** • **Cross-Lesson: 4.02 The Poisson Distribution**

The half-line $\arg(z) = 45^\circ$ intersects the circle $|z| = 4$ at which point?

a. $(2\sqrt{2}3, 2\sqrt{2})$

b. $(-2\sqrt{2}, 2\sqrt{2})$

c. $(2\sqrt{2}, -2\sqrt{2})$

d. $(2\sqrt{2}, 2\sqrt{2})$

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41 [Challenging] • Target: Angle between a line and a plane • Cross-Lesson: 5.01 Further Integration Techniques

Consider a point $P(x, y)$ that lies on the parabola defined by the equation $y = x^2 - 6x + 13$. Let F be the focus of this parabola. If the distance from P to the directrix of the parabola is minimized, what is the y-coordinate of the point P?

a. 3

b. 4

c. 5

d. 6

42 [Challenging] • Target: Distance from a point to a plane • Cross-Lesson: 5.02 Differential Equations

Consider a unit circle centered at the origin. A point $P(x, y)$ lies on the unit circle in the second quadrant such that the angle formed by the positive x-axis and the line segment OP is θ. If $\sin(\theta) = \frac{3}{5}$ and the y-coordinate of P is $y = \frac{3}{5}$, determine the x-coordinate of P.

a. $-\frac{4}{5}$

b. $\frac{4}{5}$

c. $-\frac{3}{5}$

d. $-\frac{25}{16}$

43 [Challenging] • Target: Finding the angle between two planes • Cross-Lesson: 5.03 Maclaurin Series & Convergence of Series

The graph of $y = f(x)$ is a curve passing through the points A(1, 2), B(4, 2), and C(2, 5). This graph is then subjected to a sequence of transformations. First, it is reflected in the line $y = x$. Let the image of the points A, B, and C under this reflection be A', B', and C' respectively. Secondly, the graph of the reflected curve is translated by the vector $(-3; 0)$. Let the final image of the points A', B', and C' after this translation be A'', B'', and C''. Given that the original curve is a parabola with its axis of symmetry parallel to the y-axis, and passes through A, B, and C, find the coordinates of C''.

a. (-3, 2)

b. (-2, -3)

c. (2, -3)

d. (-3, -2)

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44 [Challenging] • Target: Finding the intersection line of two planes • Cross-Lesson: 5.04 Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)

A particle's position in the xy-plane is given by the parametric equations $x(t) = 2t + 1$ and $y(t) = t^2 - 3$, where t is time. What is the Cartesian equation of the path traced by the particle?

a. $y = \frac{1}{2}x^2 - 3$

b. $y = \left(\frac{x-1}{2}\right)^2 - 3$

c. $y = x^2 - 3$

d. $y = 2\left(\frac{x-1}{2}\right) + 1$

45 [Challenging] • Target: Testing whether a line lies within a plane • Cross-Lesson: 6.01 Matrices (AI HL)

Consider the vector $\mathbf{v}$ with initial point $P = (2, 3)$ and terminal point $Q = (5, 7)$. What are the components of the vector $\mathbf{v}$?

a. ■3, 4■

b. ■7, 10■

c. ■-3, -4■

d. ■4, 3■

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